AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

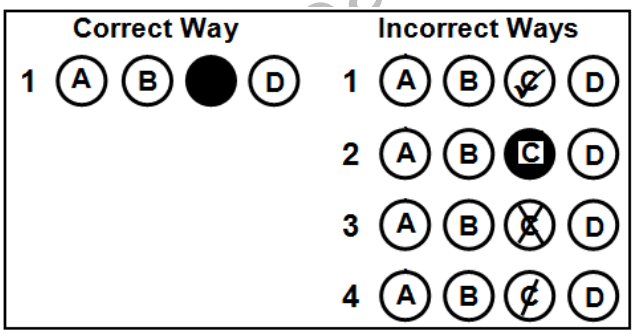
General Mathematics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The value of algebraic expression $\frac{xy^2 - x^2y}{xy}$ at $x = -1$ and $y = -1$ is

A. 0
B. 1
C. 2
D. -2

2. $x^2 - \left(\frac{1}{2}\right)^2$ is equal to

A. $\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$
B. $\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$
C. $\left(x + \frac{1}{4}\right)\left(x - \frac{1}{4}\right)$
D. $\left(x - \frac{1}{4}\right)\left(x - \frac{1}{4}\right)$

3. The rational expression $\frac{9x^2 - 4}{(3x + 2)(9x - 4)}$, in its lowest term is

A. $\frac{x}{3x + 2}$
B. $\frac{3x - 2}{9x - 4}$
C. $\frac{9x - 4}{3x + 2}$
D. $\frac{3x + 2}{9x - 4}$

4. On factorisation of $3x + 6x^2 + 9x^3$, we get

A. $3x(1 + 2x + 3x^2)$
B. $3x(1 + 3x + 6x^2)$
C. $3x(1 + 2x^2 + 3x^2)$
D. $3x(3x + 6x^2)$

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5. The factors of $x^3 - \left(\frac{1}{2}\right)^3$ are

A. $\left(x-\frac{1}{2}\right)\left(x^2+x+\frac{1}{4}\right)$

B. $\left(x-\frac{1}{2}\right)\left(x^2-x+\frac{1}{4}\right)$

C. $\left(x-\frac{1}{2}\right)\left(x^2+\frac{x}{2}+\frac{1}{4}\right)$

D. $\left(x-\frac{1}{2}\right)\left(x^2-\frac{x}{2}+\frac{1}{4}\right)$

6. If $2x^2+3x-5$ is divided by $x+1$, then the remainder is

A. -10
B. -6
C. -4
D. 0

7. The least common multiple (L.C.M.) of 1 and x^2-y^2 is

A. 1
B. $x-y$
C. $x+y$
D. x^2-y^2

8. The simplified form of $\frac{1}{x-2}+\frac{1}{(x-2)^2}$ is

A. $\frac{2}{(x-2)^2}$

B. $\frac{x+3}{x^2-4}$

C. $\frac{x+1}{(x-2)^2}$

D. $\frac{x-1}{(x-2)^2}$

9. The square root of $x^2-4a^2x+4a^4$ is

A. $\pm(x-2a)$
B. $\pm(x-2a^2)$
C. $\pm(x-2a\sqrt{x}+2a)$
D. $\pm(x-2a\sqrt{x}+2a^2)$

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10. The solution set of $\frac{x}{2} \leq \frac{1}{2}$, where $x \in W$, is

A. $\{0\}$
B. $\{1\}$
C. $\{0,1\}$
D. $\{..., -2, -1, 0, 1\}$

11. The solution set of the equation $2(x-3)+4=4$ is

A. $\{1\}$
B. $\left\{\frac{3}{2}\right\}$
C. $\{3\}$
D. $\{5\}$

12. The solution set of the equation $\sqrt{\frac{x-a}{a}} = 1$ is

A. $\{2a\}$
B. $\{\pm a^2\}$
C. $\{\pm 2a\}$
D. $\{a^2 + a\}$

13. The standard form of the quadratic equation $(x-a)(x-b)=0$ is

A. $x^2 + (a-b)x + ab = 0$
B. $x^2 - (a-b)x - ab = 0$
C. $x^2 - (a+b)x - ab = 0$
D. $x^2 - (a+b)x + ab = 0$

14. The root(s) of the quadratic equation $x^2 - \frac{x}{3} = 0$ is/ are

A. $\left\{\frac{1}{3}\right\}$
B. $\left\{-\frac{1}{3}\right\}$
C. $\left\{0, \frac{1}{3}\right\}$
D. $\left\{0, -\frac{1}{3}\right\}$

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15. If $A = \begin{bmatrix} 3 & 4 \\ a & 8 \end{bmatrix}$ and $|A| = 0$, then a is equal to

A. -7
B. -6
C. 6
D. 7

16. Which of the following is/ are symmetric matrix/ matrices?

$$X = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & 0 & a \\ 0 & b & 0 \\ c & 0 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 0 & 0 & a \\ 0 & a & 0 \\ a & 0 & 0 \end{bmatrix}$$

A. X only
B. Y only
C. X and Z
D. Y and Z

17. If the determinant of $\begin{bmatrix} 5 & a \\ 6 & a \end{bmatrix}$ is $a - 2$, then value of a will be

A. $-\frac{1}{5}$
B. 0
C. 1
D. 2

18. The matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is a

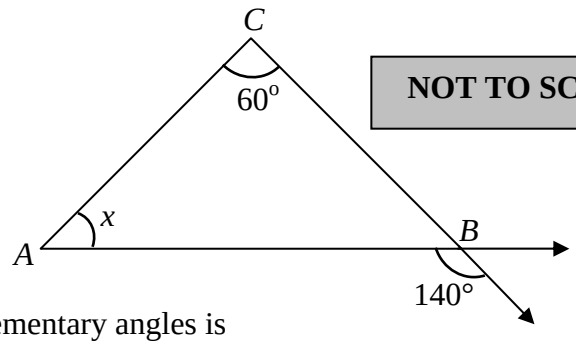
A. scalar matrix.
B. square matrix.
C. identity matrix.
D. diagonal matrix.

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19. Concentric circles have the same

A. radii.
B. areas.
C. centres.
D. tangents.

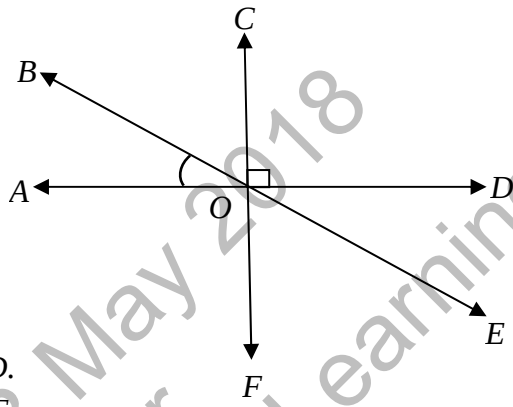
20. In the given diagram, the value of x is



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A. 40°
B. 60°
C. 80°
D. 140°

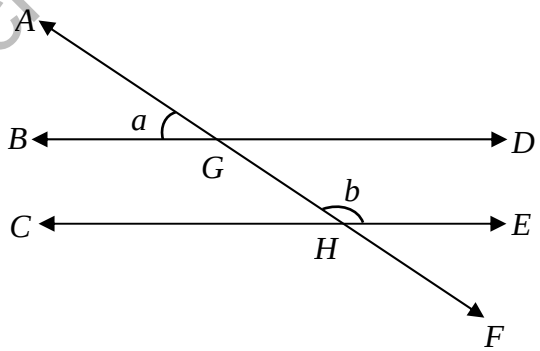
21. In the given diagram, one pair of supplementary angles is



A. $\angle AOB$ and $\angle BOD$.
B. $\angle DOE$ and $\angle EOF$.
C. $\angle COD$ and $\angle BOC$.
D. $\angle AOB$ and $\angle DOE$.

22. In the given diagram, if BD and CE are parallel lines then, $\angle AGB + \angle DGA$ is equal to

I. 180°
II. $a + b$
III. $a + 90^o$



A. I only
B. I and II
C. I and III
D. II and III

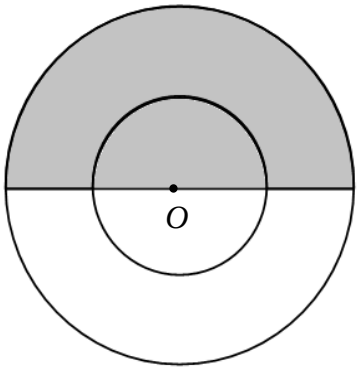
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23. The three medians of any triangle are

I. concurrent.
II. of same length.
III. same as the right bisector.

A. I only
B. II only
C. I and III
D. II and III

24. In the given diagram, two concentric circles are shown with centre *O*. If *a* units and *b* units are radii of the inner and outer circles respectively, then the area of the shaded region is



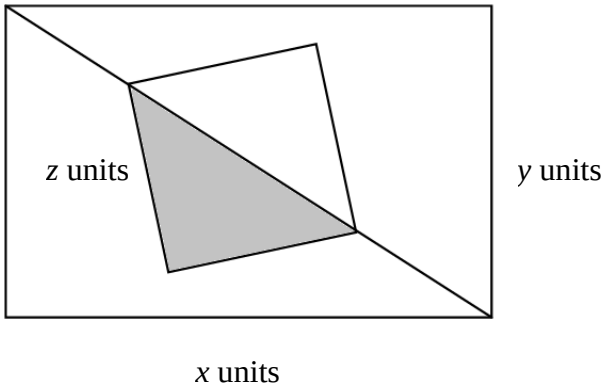
A. $\pi\left(\frac{b}{2}\right)^2$ units^2

B. $\frac{\pi b^2}{2}$ units^2

C. $\frac{\left(\pi b^2 + \pi a^2\right)}{2}$ units^2

D. $\pi\left(\frac{b}{2}\right)^2 + \pi\left(\frac{a}{2}\right)^2$ units^2

25. The diagram shows a square and a rectangle. The length and breadth of the rectangle are *x* units and *y* units respectively, and length of the side of the square is *z* units. The area of the shaded region is



A. $\frac{xy}{4}$ units^2

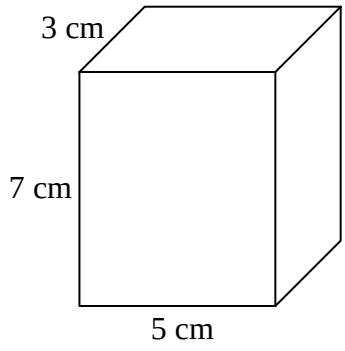
B. $\frac{z^2}{2}$ units^2

C. $\frac{\sqrt{x^2 + y^2}}{2}$ units^2

D. $xy - \frac{z^2}{2}$ units^2

26. The diagram shows a cuboid and the lengths of its different sides. The volume of the cuboid is

A. 15 cm
B. 15 cm^3
C. 105 cm
D. 105 cm^3



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27. A right circular cylinder has a height h cm and its base has a diameter d cm. The volume of the circular cylinder is

A. $\pi d^2 h \text{ cm}^3$

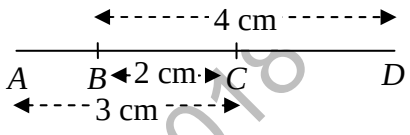
B. $\pi \frac{d^2}{2} h \text{ cm}^3$

C. $\pi \frac{d^2}{4} h \text{ cm}^3$

D. $\pi \left(\frac{d}{4}\right)^2 h \text{ cm}^3$

28. In the given figure A, B, C and D are collinear points. If $AC = 3$ cm, $BD = 4$ cm and $BC = 2$ cm, then AD is

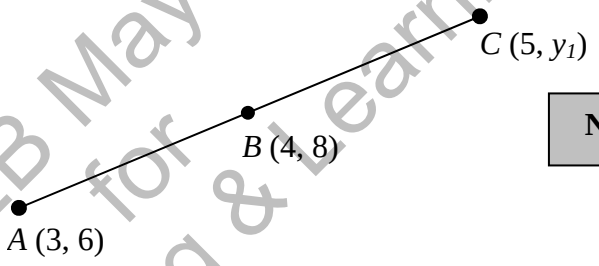
A. 5 cm
B. 6 cm
C. 7 cm
D. 8 cm



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29. In the given diagram, if B is the midpoint of line segment AC, then the y-coordinate of point C is

A. 2
B. 7
C. 10
D. 13



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30. If the distance between points (a, a) and $(0, 0)$ is $\sqrt{2}$, then one of the values of a is

A. $\frac{1}{\sqrt{2}}$
B. 1
C. $\sqrt{2}$
D. 2

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